

# Envelope Theorems

Let  $X$  be a nonempty subset of  $\mathbb{R}^n$  for some positive integer  $n$ , and let  $T$  be nonempty open real interval, and let  $f : X \times T \rightarrow \mathbb{R}$ .

## 1 Constraint set depends on $t$

Set  $X = \mathbb{R}^n$ , and let  $g : \mathbb{R}^n \times T \rightarrow \mathbb{R}^m$  for some positive integer  $m$ . Consider the problem

$$V(t) = \max_{x \in \mathbb{R}^n} f(x, t) \tag{1}$$

subject to the constraint that  $g(x, t) \leq 0$ . (Any nonnegativity constraints on  $x$  are included in the vector of functions  $g$ .) Let  $\mathcal{L}(x, \lambda, t) = f(x, t) - \lambda \cdot g(x, t)$  and  $x^*(t) = \operatorname{argmax}_{x \in X} f(x, t)$  subject to  $g(x, t) \leq 0$ , assumed *nonempty* on  $T$ .

**Theorem** [Classical Envelope Theorem] Assume that

- $x^*(\cdot)$ ,  $f$ , and  $g$  are each  $C^1$  functions on their domains.<sup>1</sup>
- The Kuhn-Tucker *necessary* conditions for the problem hold at  $(x^*(t), t)$  for each  $t \in T$  with associated multipliers  $\lambda^*(t) = (\lambda_1^*(t), \dots, \lambda_m^*(t)) \geq 0$ ; and  $\lambda^*(\cdot)$  is also a  $C^1$  function on  $T$ .

Then  $V$  is differentiable at any  $t_0 \in T$  and

$$\boxed{V'(t_0) = f_t(x^*(t_0), t_0) - \sum_{i=1}^m \lambda_i^*(t_0) g_i^t(x^*(t_0), t_0),} \quad \star$$

where  $g = (g^1, \dots, g^m)$ . More compactly,

$$\boxed{V'(t_0) = \mathcal{L}_t(x^*(t_0), \lambda^*(t_0), t_0).}$$

*Proof:* Differentiate  $V(t) = f(x^*(t), t) - \lambda^*(t) \cdot g(x^*(t), t)$  (why is the equality true?) with respect to  $t$ , evaluate it at  $t = t_0$ , and use the Kuhn-Tucker conditions.  $\square$

In words: under the two sets of assumptions, the value function is differentiable and we can calculate it simply by differentiating the Lagrangean  $\mathcal{L}(x, \lambda, t)$  with respect to  $t$  *before optimizing*, then evaluate the derivative at the point  $(x, \lambda, t) = (x^*(t_0), \lambda^*(t_0), t_0)$ .

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<sup>1</sup>In particular,  $x^*(\cdot)$  is single-valued and differentiable on  $T$ .

## 2 Constraint set does not depend on $t$

Now let  $X$  be an arbitrary nonempty set. Assume that, for every  $t \in T \subset \mathbb{R}$ , there is a maximizer of  $f(\cdot, t)$  on  $X$ .

**Theorem** (Milgrom and Segal, 2002): If for every  $x \in X$ ,  $f(x, \cdot)$  is differentiable on  $T$  and if  $V(t) = \max_{x \in X} f(x, t)$  is differentiable at  $t_0 \in T$ , then

$$\boxed{V'(t_0) = f_t(x_0, t_0),}$$

for *any* maximizer  $x_0$  of  $f(\cdot, t_0)$  on  $X$ .<sup>2</sup>

*Proof:* Let  $x_0$  be any maximizer of  $f(\cdot, t_0)$  on  $X$ . Consider any sequence  $t_k$  in  $T$  converging to  $t_0$  from above. Since  $V(t_k) \geq f(x_0, t_k)$  and  $t_k \geq t_0$ , we have

$$\frac{V(t_k) - V(t_0)}{t_k - t_0} \geq \frac{f(x_0, t_k) - f(x_0, t_0)}{t_k - t_0}. \quad (2)$$

Take the limit as  $k \rightarrow \infty$  to find that the right derivative satisfies  $V'(t_0^+) \geq f_t(x_0, t_0)$ . Now consider any sequence  $t_k$  converging to  $t_0$  from below. The inequality (2) is reversed, since the denominator changes sign. Take the limit as  $k \rightarrow \infty$  to find that the left derivative  $V'(t_0^-) \leq f_t(x_0, t_0)$ . Since  $V$  is differentiable at  $t_0$ , the left and right derivatives must agree and equal  $V'(t_0)$ , so that  $V'(t_0) = f_t(x_0, t_0)$ .  $\square$

**Remarks:**

- If  $V$  is differentiable at  $t_0$  and  $f_t(\cdot, t_0)$  is one-to-one, then the maximizer of  $f$  on  $X$  is unique at  $t_0$ . For example, if the profit function  $\pi(p) = \max_{y \in Y} p \cdot y$  is differentiable at  $p$ , then the profit-maximizing production plan at  $p$  is unique.
- *Saddlepoint inequalities.* Consider problem (1). If there is a  $\lambda^* \in \mathbb{R}_+^m$  and  $x^* \in \mathbb{R}^n$  such that  $x^*$  maximizes  $\mathcal{L}(x, \lambda^*, t)$  on  $\mathbb{R}^n$ ; and  $\lambda^*$  minimizes  $\mathcal{L}(x^*, \lambda, t)$  on  $\mathbb{R}_+^m$ , then  $(x^*, \lambda^*)$  solves the *saddlepoint inequalities*

$$\mathcal{L}(x, \lambda^*, t) \leq \mathcal{L}(x^*, \lambda^*, t) \leq \mathcal{L}(x^*, \lambda, t) \quad \text{for all } (x, \lambda) \in \mathbb{R}^n \times \mathbb{R}_+^m. \quad (3)$$

Note that the max problem (the first inequality) is an *unconstrained* problem. If (3) holds, it is easy to show that  $x^*$  solves the constrained problem (1). Under some conditions,<sup>3</sup> the converse holds: if  $x^*$  solves (1), then there is a  $\lambda^* \in \mathbb{R}_+^m$  such that  $(x^*, \lambda^*)$  satisfies the saddlepoint inequalities (3). In that case we can relax the assumptions made in Section 1 that the functions  $(x^*(\cdot), \lambda^*(\cdot))$  are differentiable in  $t$  and  $f, g$  are differentiable in  $x$ . (Use the first inequality on page 595 of Milgrom and Segal, 2002, and take limits as  $k \rightarrow \infty$ .)

<sup>2</sup>This is Theorem 1 in Milgrom, P. and I. Segal, “Envelope Theorems for Arbitrary Choice Sets,” *Econometrica* 70 (2002): 583-602. Sinander, “The Converse Envelope Theorem” (2021), working paper, substantially generalizes the Milgrom-Segal theorem.

<sup>3</sup>For example,  $f$  is concave, the constraint functions are all convex, and *Slater’s* constraint qualification is met: there is a point  $\hat{x}$  in  $\mathbb{R}^n$  with  $g(\hat{x}) < 0$ .